

Hypothesis Testing

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Introduction

Test of Hypothesis - Overview

- A statistical hypothesis is a claim about a population parameter (population mean, population proportion, population variance etc.)
- We test this claim using sample data
- The goal is to make a decision under uncertainty
- Decisions are made using probability and sampling distributions

Statistical Hypotheses

- **Null hypothesis (H_0):** baseline assumption
- **Alternative hypothesis (H_A):** competing claim
- They are mutually exclusive and exhaustive
- Example:

$$H_0 : \mu = 100 \quad \text{vs} \quad H_A : \mu \neq 100$$

- We use sample to assess whether the data provides any evidence *against* the null hypothesis
- Based on the available data, we either reject the null hypothesis in favor of the alternative or “fail to reject the null hypothesis”
- Therefore, we do not “accept” the null hypothesis

Types of Tests (Alternative Hypotheses)

- Two-tailed:

$$H_A : \mu \neq \mu_0$$

- Left-tailed:

$$H_A : \mu < \mu_0$$

- Right-tailed:

$$H_A : \mu > \mu_0$$

Types of Errors

Type I Error

- Rejecting H_0 when it is true
- Probability of type I error (also known as *level of significance*):

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ true})$$

- Also called false positive

Type II Error

- Failing to reject H_0 when it is false
- Probability of type II error:

$$\beta = P(\text{Fail to reject } H_0 \mid H_0 \text{ false})$$

- Also called false negative

Errors Summary Table

	H_0 True	H_0 False
Reject H_0	Type I	Correct
Fail to reject H_0	Correct	Type II

Level of Significance

- $\alpha = P(\text{Type I error}) = P(\text{Reject } H_0 \mid H_0 \text{ true})$
- Common choices: 0.10, 0.05, 0.01
- It is essentially our “tolerance” for type I error
- Smaller choice of α means stricter evidence required

Choosing α in Practice

- There is no universally “correct” α — the choice should reflect how costly a Type I error is in the given context
- Use a **stricter (smaller)** α (e.g. 0.01) when a false positive is dangerous or expensive
 - Example: approving a new drug as safe when it is not; certifying a structural design as safe when it is not
- Use a **looser (larger)** α (e.g. 0.10) when missing a real effect is costlier than a false alarm
 - Example: early - stage screening for a promising material or drug candidate
- $\alpha = 0.05$ is the conventional default when there is no strong reason to move away from it

Power of a Test

- **Power** is the probability of correctly rejecting H_0 when it is actually false

$$\text{Power} = 1 - \beta = P(\text{Reject } H_0 \mid H_0 \text{ false})$$

- A test with low power may fail to reject a false H_0 , a *false negative* error
- Power depends on:
 - **Sample size** n : larger $n \Rightarrow$ higher power
 - **Effect size** $|\mu - \mu_0|$: larger effect $\Rightarrow$ easier to detect
 - **Variability** σ : smaller $\sigma \Rightarrow$ higher power
 - **Significance level** α : larger $\alpha \Rightarrow$ higher power (but at the cost of higher Type I error probability)
- Conventionally, power ≥ 0.80 is considered adequate

Choosing Sample Size via Power

- Sample size should be decided **before** data collection, not after
- Given a target power (typical convention is to set power at 0.80), the minimum required n can be calculated
- Larger n costs more time and resources, but too small an n can lead to too low power (a high β)
- In practice, this calculation is done using statistical software (power analysis), not by hand

Test Outcomes and Their Probabilities

	H_0 True	H_0 False
Reject H_0	Type I (α)	Correct ($1 - \beta$)
Fail to reject H_0	Correct ($1 - \alpha$)	Type II (β)

- α = level of significance
- $1 - \beta$ = power of a test

Test about a Single Mean

Test about a Single Mean — Test Statistic

- Used to test a claim about a population mean μ based on a sample mean $\bar{x}$

- **Test statistic:**

$$t_{\text{cal}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \quad \text{under } H_0$$

- The t -test is used regardless of sample size — it converges to the Z -test for large n , but remains valid and more robust for small n

Test about a Single Mean — Example

Problem: It is claimed that the mean weight of teens in a city is 35 kg. A sample of $n = 10$ teens yields $\bar{x} = 40.1$ kg, $s = 4.2$ kg. At $\alpha = 0.05$, does the mean weight differ from 35 kg?

Let μ = mean weight of the teens.

$$H_0 : \mu = 35 \quad \text{vs} \quad H_A : \mu \neq 35$$

Assumptions: Population is normally distributed; σ unknown.

Test about a Single Mean — Solution

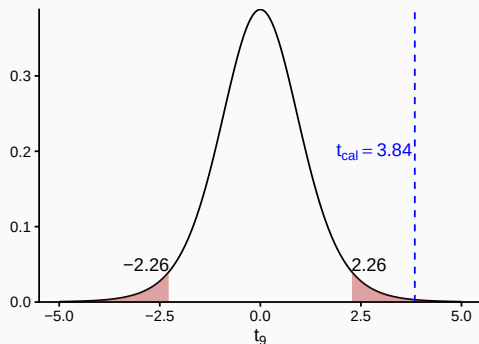
Test statistic:

$$t_{\text{cal}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{40.1 - 35}{4.2/\sqrt{10}} = 3.84$$

Critical value: $t_{0.025, 9} = 2.262$

Decision: Since $t_{\text{cal}} = 3.84 > 2.262$, we **reject** H_0 at 5% level of significance.

There is strong evidence that the mean weight differs from 35 kg.



Test about Two Means (Independent Samples)

Two Means (Independent, Equal Variances) — Test Statistic

- Used to compare the means of two **independent** populations, assuming equal population variances ($\sigma_1^2 = \sigma_2^2$)
- **Pooled variance:**

$$S^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

- **Test statistic:**

$$t_{\text{cal}} = \frac{\bar{x}_1 - \bar{x}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1 + n_2 - 2} \quad \text{under } H_0$$

Two Means (Independent, Equal Variances) — Example

Problem: A sample of 9 bulbs from Company A had a mean lifetime of 1309 hrs ($s_1 = 419$); a sample of 16 bulbs from Company B had a mean of 1205 hrs ($s_2 = 390$). Do the two companies' bulbs differ significantly in mean lifetime? Use $\alpha = 0.05$. Assume **equal** population variances.

Let μ_1, μ_2 = mean lifetimes of bulbs from Company A and B respectively.

$$H_0 : \mu_1 = \mu_2 \quad \text{vs} \quad H_A : \mu_1 \neq \mu_2$$

Two Means (Independent, Equal Variances) — Solution

Pooled variance:

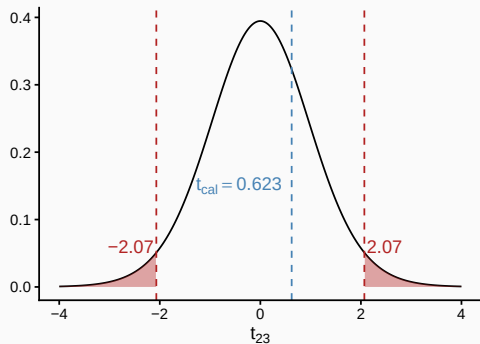
$$S^2 = \frac{8 \times 419^2 + 15 \times 390^2}{23} = 160260.3$$

Test statistic:

$$t_{\text{cal}} = \frac{1309 - 1205}{S \sqrt{\frac{1}{9} + \frac{1}{16}}} = 0.623$$

Critical value: $t_{0.025, 23} = 2.069$

Decision: Since $|t_{\text{cal}}| = 0.623 < 2.069$, we **fail to reject** H_0 at 5% level of significance. There is no significant difference between the mean lifetimes.



Test about Two Means (Dependent Samples)

Test about Two Means — Dependent (Paired) Samples

- Used when the two samples are **not independent**
- Common when the same subjects are measured **before and after** a treatment, or under two conditions
- Define the **difference**: $d_i = x_i - y_i$
- This reduces the two-sample problem to a **one-sample** t -test on the differences

Test statistic:

$$t_{\text{cal}} = \frac{\bar{d}}{s_d/\sqrt{n}} \sim t_{n-1} \quad \text{under } H_0$$

where $\bar{d} = \frac{\sum d_i}{n}$ and $s_d^2 = \frac{1}{n-1} \left(\sum d_i^2 - \frac{(\sum d_i)^2}{n} \right)$

Two Means (Dependent) — Example

Problem: Nine adults agreed to test a new diet program. Their weights (in lb) were recorded before and after the program. Test whether the program is effective in reducing weight at $\alpha = 0.05$.

	Adult								
	1	2	3	4	5	6	7	8	9
Before (x_i)	132	138	126	114	122	132	142	119	137
After (y_i)	130	132	130	111	122	133	139	119	130

$$H_0 : \mu_d = 0 \quad \text{vs} \quad H_A : \mu_d > 0$$

Two Means (Dependent) — Solution

Summary statistics:

$$\bar{d} = \frac{4}{9} = 0.44, \quad s_d^2 = 21.53$$

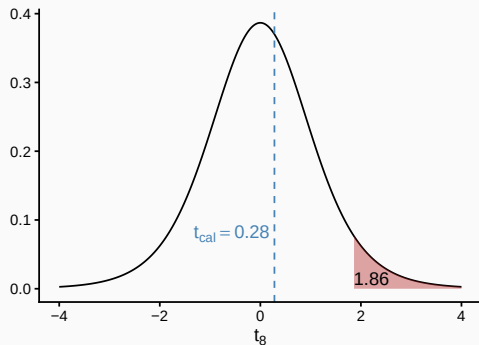
Test statistic:

$$t_{\text{cal}} = \frac{0.44}{4.64/\sqrt{9}} = 0.28$$

Critical value: $t_{0.05, 8} = 1.86$

Decision: Since $t_{\text{cal}} = 0.28 < 1.86$, we **fail to reject** H_0 at 5% level of significance.

Conclusion: There is insufficient evidence to conclude that the new diet program reduces weight.



P - value

P-value: Meaning and Decision Rule

- The **p-value** is the probability of observing a test statistic as extreme as, or more extreme than, the one calculated from the sample — *assuming H_0 is true*
- It is **not** the probability that H_0 is true
- A **small** p-value means the observed data would be very unlikely if H_0 were true — evidence against H_0
- A **large** p-value means the data are consistent with H_0 , but this does *not* prove H_0 is true
- Decision rule:
 - $p \leq \alpha$: reject H_0
 - $p > \alpha$: fail to reject H_0
- P-value is reported statistical software, and usually not calculated by hand

Thank you.
